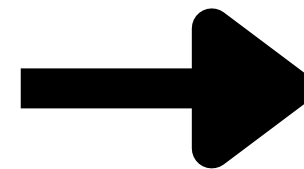


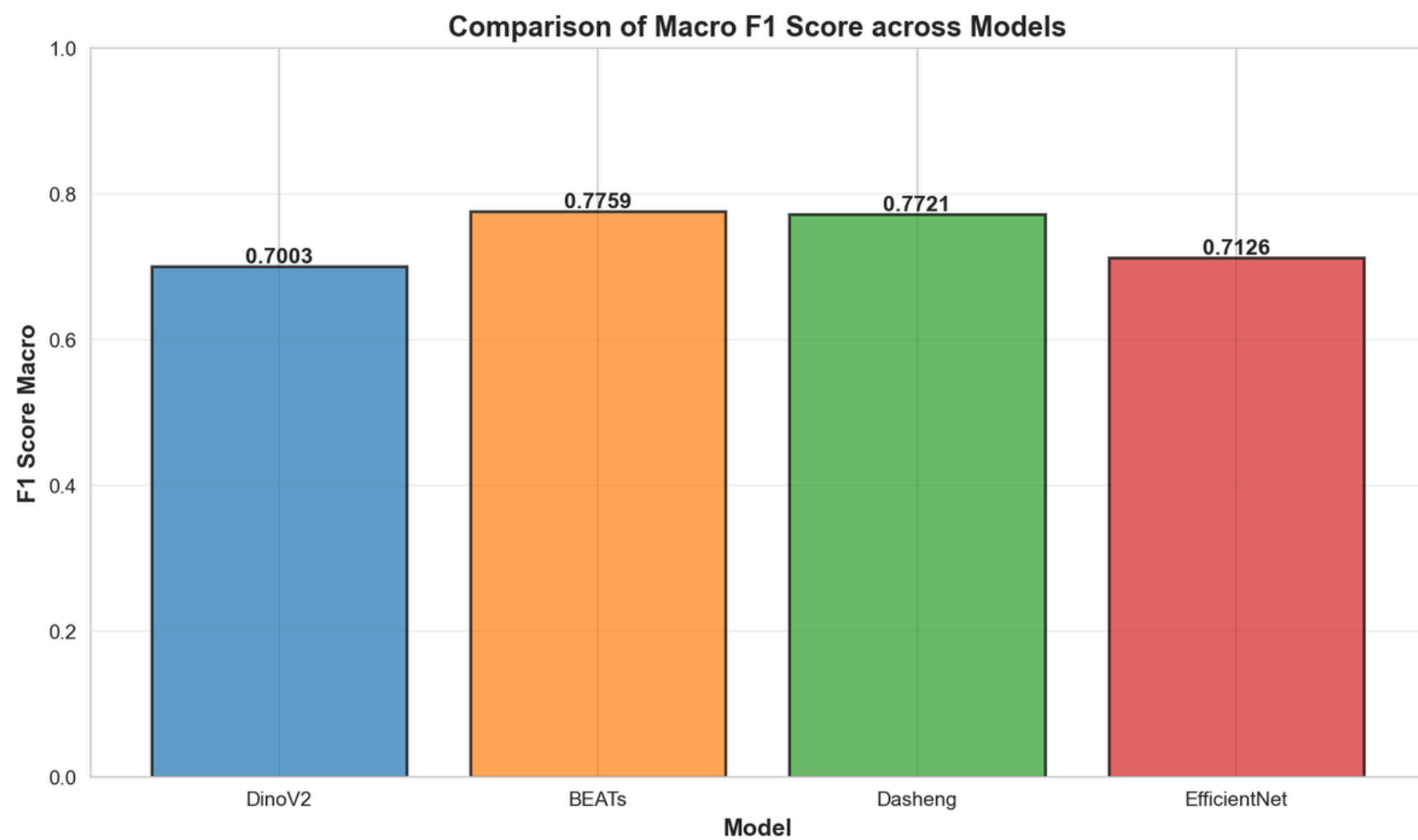
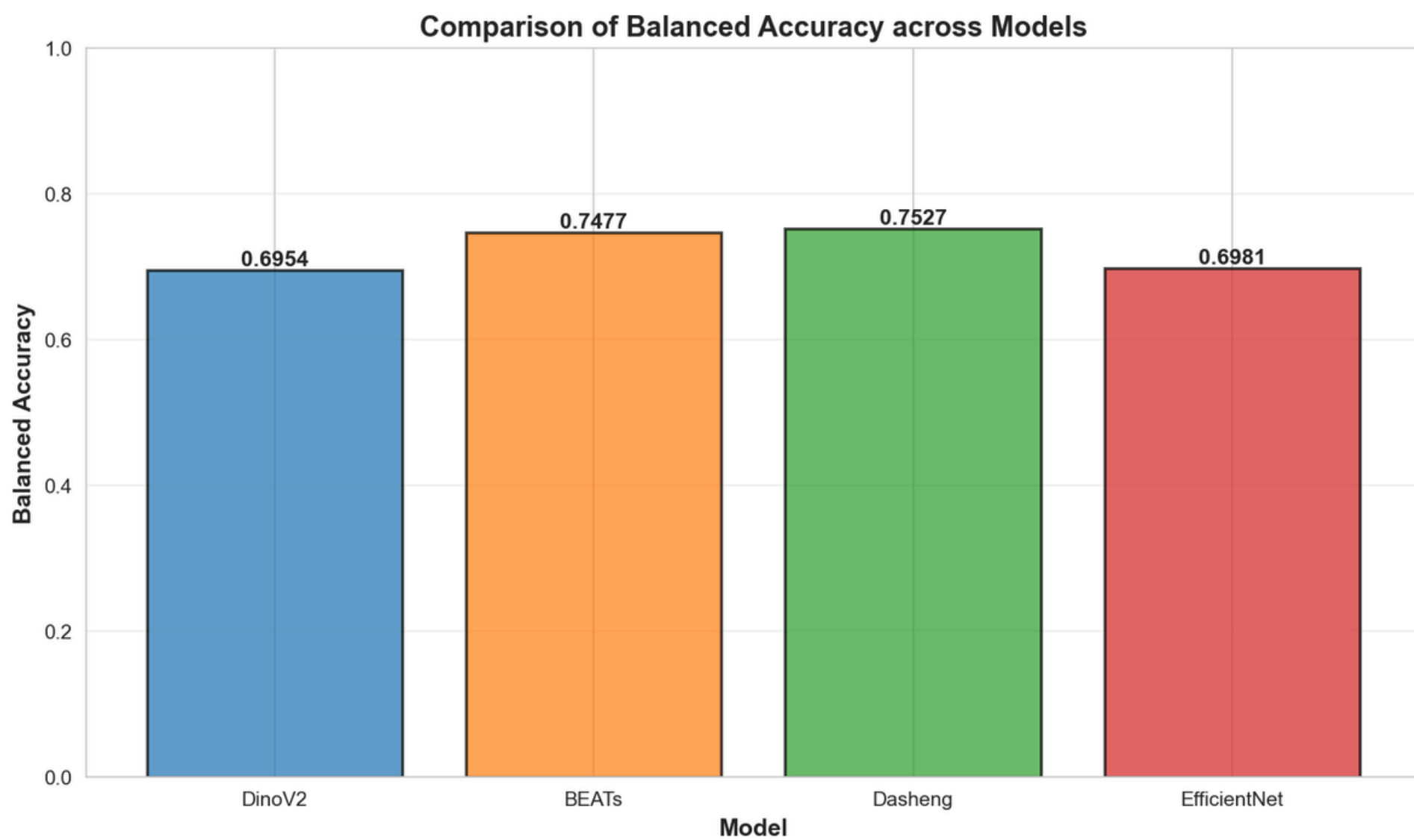
Benchmark modèles d'autosupervision pré-entraînés pour calculs d'embeddings sur données audios (audios bruts ou mel-spectrogrammes)

- DinoV2 Large (dinov2_vitl14)
- EfficientNet (EfficientNet-B0)
- VGG16
- BEATs (BEATs_iter3_plus_AS2M)
- Dasheng (base)

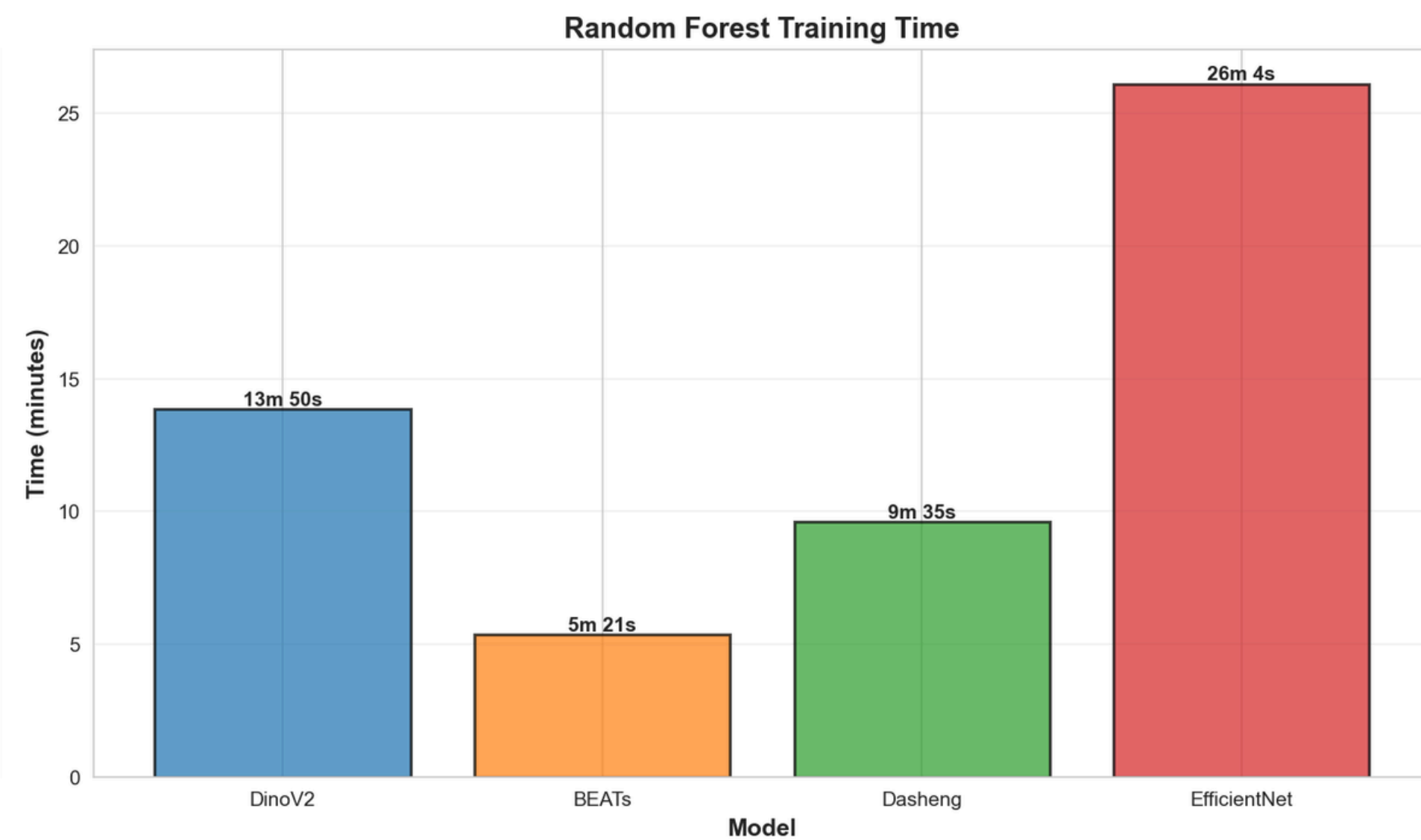
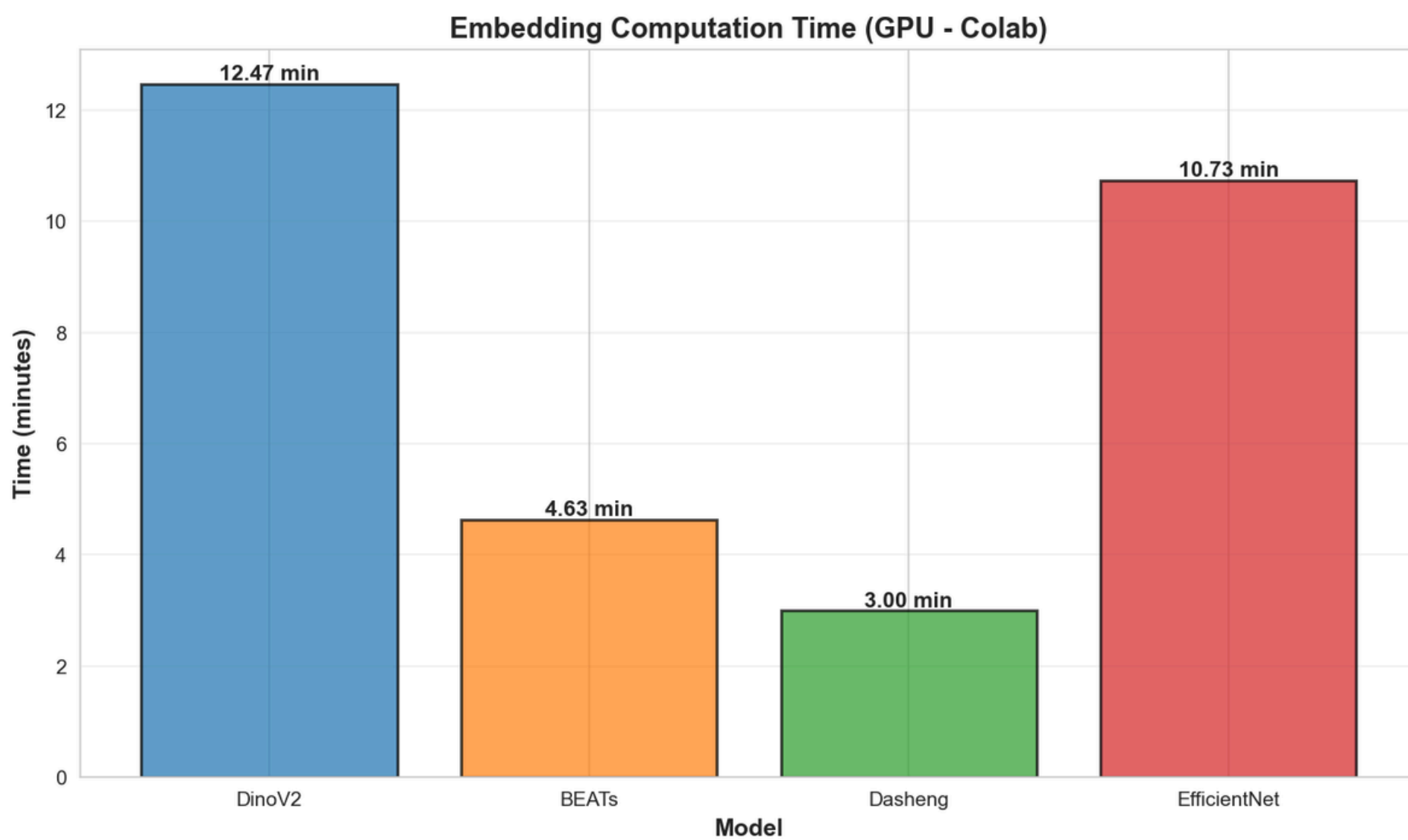


Mel-spectrogramme
Mel-spectrogramme
Mel-spectrogramme
Audio brut
Audio brut

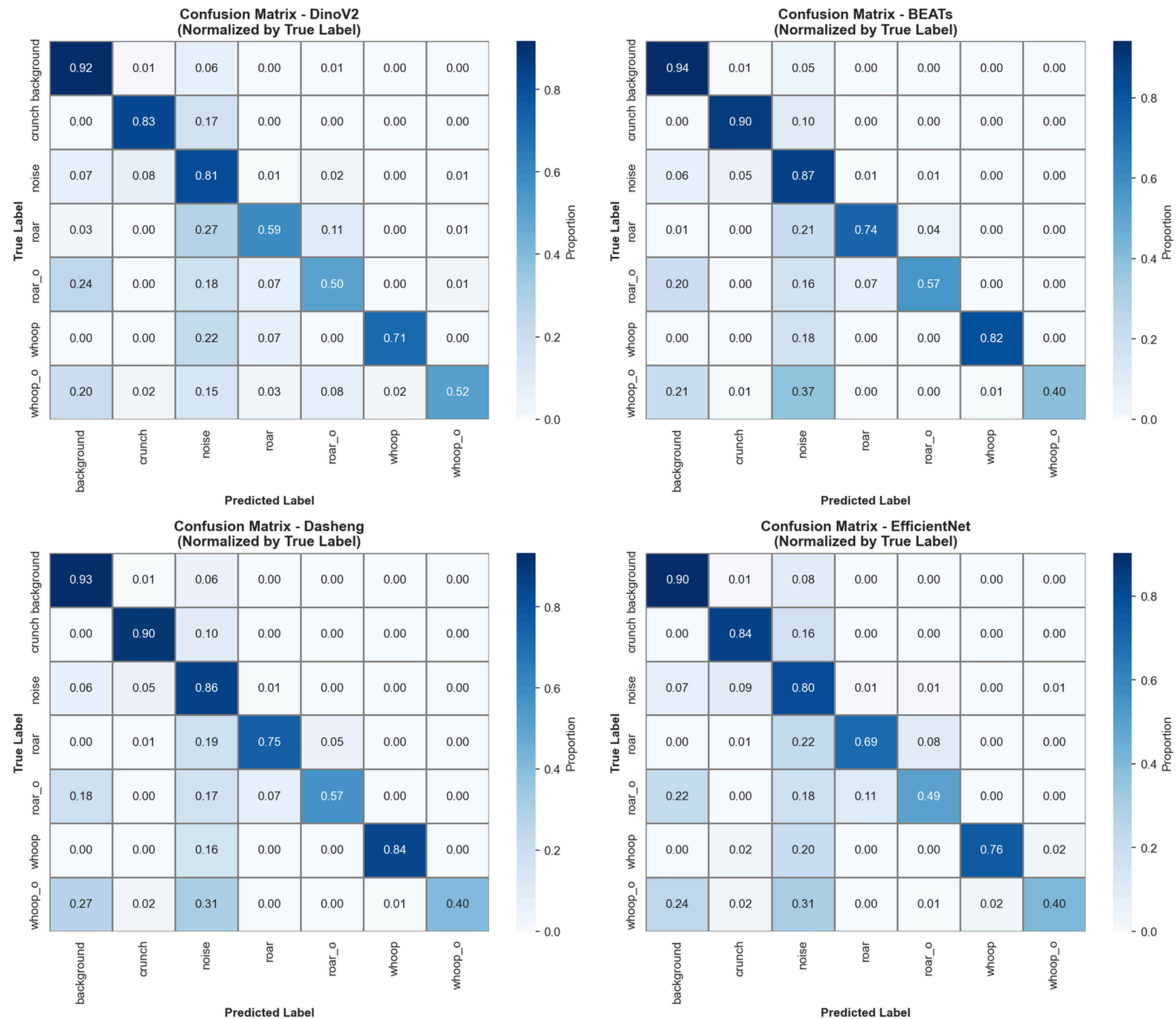
Métriques de performance



Métriques de temps de calculs



Matrices de confusion



Valeurs d'AUC

Model	Macro-AUC	background_AUC	crunch_AUC	noise_AUC	roar_AUC	roar_o_AUC	whoop_AUC	whoop_o_AUC
DinoV2	0.8307	0.9294	0.8911	0.8495	0.7907	0.7420	0.8547	0.7575
BEATs	0.8615	0.9447	0.9334	0.8935	0.8682	0.7823	0.9107	0.6976
Dasheng	0.8636	0.9404	0.9345	0.8894	0.8734	0.7844	0.9216	0.7013
EfficientNet	0.8313	0.9213	0.8934	0.8418	0.8426	0.7421	0.8770	0.7011

Analyse des performances par classes



Comparaison multicritères

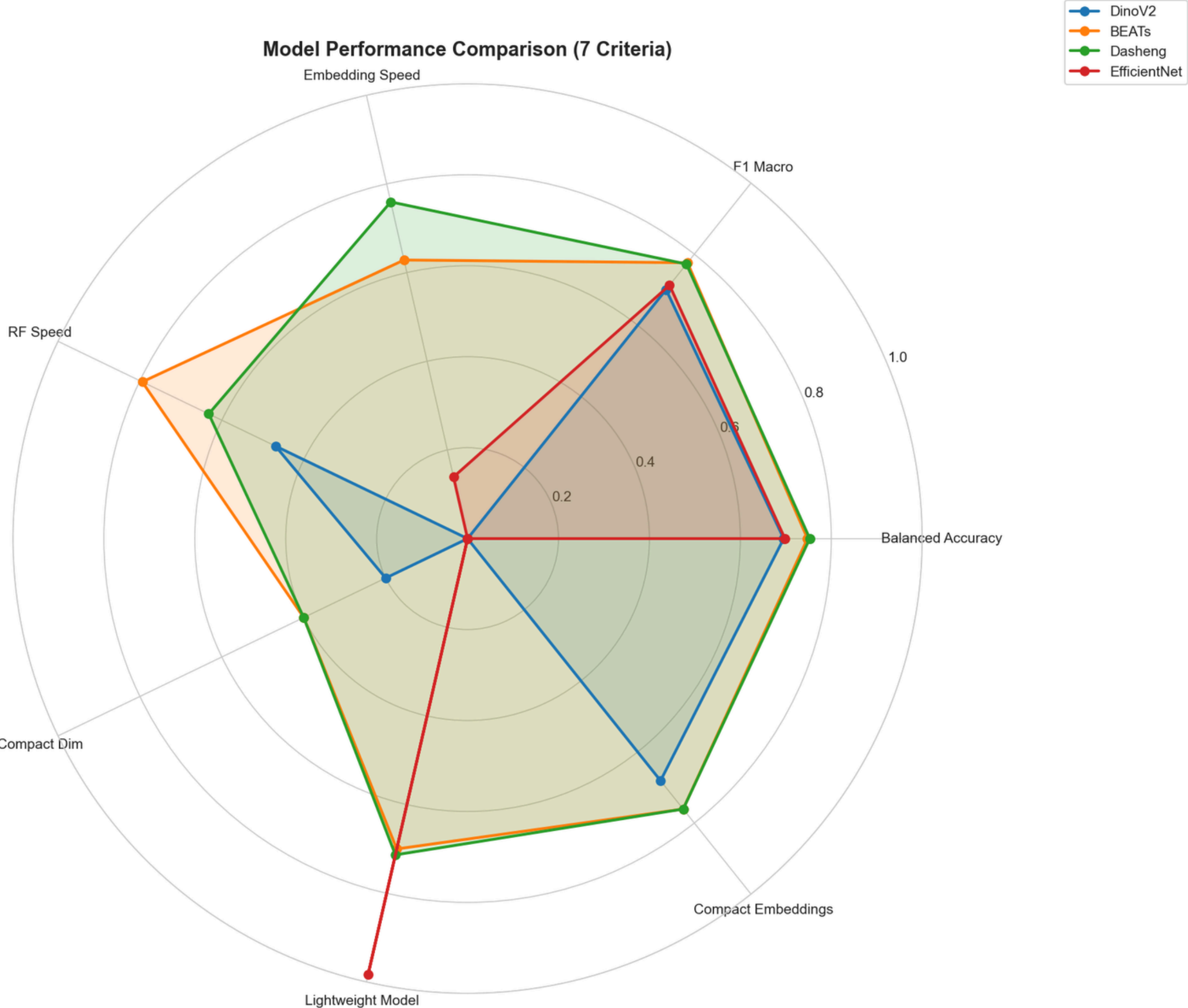


Tableau récapitulatif

	Balanced Accuracy	F1 Score Macro	Embedding Dim	Model Params (M)	Embed Time (s)	Embed Time (min)	RF Training (s)	RF Training (min)	Embeddings (MB)
DinoV2	0.6954	0.7003	1024	300.00	748	12.47	831	13.85	60.0
BEATs	0.7477	0.7759	768	90.00	278	4.63	322	5.37	45.0
Dasheng	0.7527	0.7721	768	86.00	180	3.00	576	9.60	45.0
EfficientNet	0.6981	0.7126	1280	5.27	644	10.73	1565	26.08	188.0